

Theophilus Ofori Agyekum

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EDUCATION

Kwame University of Science and Technology

BSc Chemistry

Second Class Honors (Upper Division)

Kumasi, Ghana

November 2023

EXPERIENCE

Christ Apostolic School Complex

Mathematics and Science Teacher

Accra, Ghana

September 2025 – Present

- Instructing students in core educational principles, concepts, and in-depth subject matter
- Developing and implementing engaging lesson plans tailored to diverse learning styles and academic needs
- Providing individualized support and feedback to students
- Utilizing various educational technologies and resources to enhance the learning experience and facilitate interactive lessons

Generation Ghana

Freelance Data Analytics Trainee

Accra, Ghana

April 2025 – Present

- Leveraging Python and SQL to clean, process and transform datasets of over 100,000 rows
- Delivering actionable insights to help clients optimize sales strategies and reduce operational costs up to 18%
- Collaborating with clients to deliver tailored data driven solutions to improve operational efficiency by 20% minimum
- Identified key business trends for a retail client, resulting in a 15% increase in monthly revenue within three months

Kumasi Technical University

Research Assistant

Kumasi, Ghana

October 2023 – October 2024

- Engaged in research to explore and develop the best method for the production of fibers from waste pineapple leaves in an environmentally safe way
- Assisted eight senior-year students in their research theses to analyze data, and provide solutions to problems encountered in chemical research
- Operated and maintained laboratory equipment
- Assisted in designing and conducting experiments

Kumasi Technical University

Teaching Assistant

Kumasi, Ghana

October 2023 – October 2024

- Performed administrative activities such as taking attendance, marking assignments, and maintaining student records
- Provided academic guidance and mentorship to students
- Set up lecture hall to facilitate lecturing
- Supervised undergraduate laboratory sessions

INDEPENDENT PROJECTS

Kwame Nkrumah University of Science and Technology

Kumasi, Ghana

Undergraduate Thesis

January 2023 – July 2023

- Designed research methodology, synthesized and characterized the free ligand and its Manganese(II) complex using spectroscopic and analytical techniques to assess their catalytic potential for sustainable polymerization
- Wrote a research-based thesis on Synthesis and Characterization of Free and Manganese(II) Complex of N,N'-Bis(Salicylidene)ethylenediamine for Ring-Opening Polymerization of Lactide

Student Project

May 2022 – August 2022

- Worked on a mini-project review on metal complexes of pyrazole derivatives as an anti-cancer agent

PUBLICATIONS

- Darmey, J., Narra, S., Achaw, O.W., Stinner, W., Ahiekpor, J.C., Ansah, H.F., N'guessan, B.A., **Agyekum, T.O.** & Nutakor, E.M.K. (2025). [A Review of Pretreatment Strategies for Anaerobic Digestion: Unlocking the Biogas Generation Potential of Wastes in Ghana](#). In *Waste* (Vol. 3, No. 3, p. 24). MDPI.
- Antwi, D. M. B., Amankwaa, I. K., Johannes, A., Agyekum, E., Amaniampong, D., **Agyekum, T. O.**, & Boateng, H. K. (2024). [A survey on the need to incorporate rainwater harvesting systems as an alternative domestic water source in Ghana](#). *Journal of Applied Science, Engineering, Technology and Management*, 2(2), 15-24.

CONFERENCES

Institute of Distance Learning & Continuing Education, KsTU, Kumasi

October 2024

- Topic: Guidelines and Overview of Contemporary Literature Review Methods Using VoSview and R

23rd Annual National Conference (Ghana Students' Chemical Society)

June 2023

- Theme: Developing the Ghanaian Chemist as a High-End Resource to Foster Research, Innovation and Industrialization

LEADERSHIP AND VOLUNTEERISM

Kwame Nkrumah University of Science and Technology

Kumasi, Ghana

Sports and Entertainment Head, Dept. of Chemistry

2021- 2023

- Certificate of Leadership for Outstanding Student Leader during the 2022-2023 academic year
- Coordinated team activities, delegated tasks, and managed resources
- Organized annual departmental sporting games for the 2021-2022 and 2022-2023 academic year
- Chaired the sports committee for the 23rd Annual National Conference of the Ghana Students' Chemical Society games in June 2023

Grow Girls in STEM

Accra, Ghana

STEM Outreach Volunteer

October 2025

- Assisted in delivering hands-on science workshops to inspire girls' interest in STEM fields
- Encouraged student participation and provided mentorship during small-group activities.

SKILLS, LANGUAGE AND INTEREST

Technical and Computational: Microsoft Office Suite, Basic programming (Python, HTML and CSS), SQL, Power BI, STATA and SPSS, ChemDraw

Language: English (Read, Write and Peer review), **French** (elementary)

Interests: Community involvement, Reading and Research, Data analysis, Soccer